

Unit 12, Lesson 1: Displaying Univariate Data



Benchmarks

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.

MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical, whether it is univariate or bivariate and interpret the different components and quantities in the display.



Learning Targets

- ☐ I can understand how various displays can be used to represent a univariate categorical data set.
- ☐ I can select an appropriate data display to represent numerical or categorical univariate data.



12.1.1 Warm-Up: Bell Work

To choose an appropriate graphical representation for data, the first step is to determine if the data is **numerical** (data values are numbers) or **categorical** (data values fall within categories).

1. Complete the table.

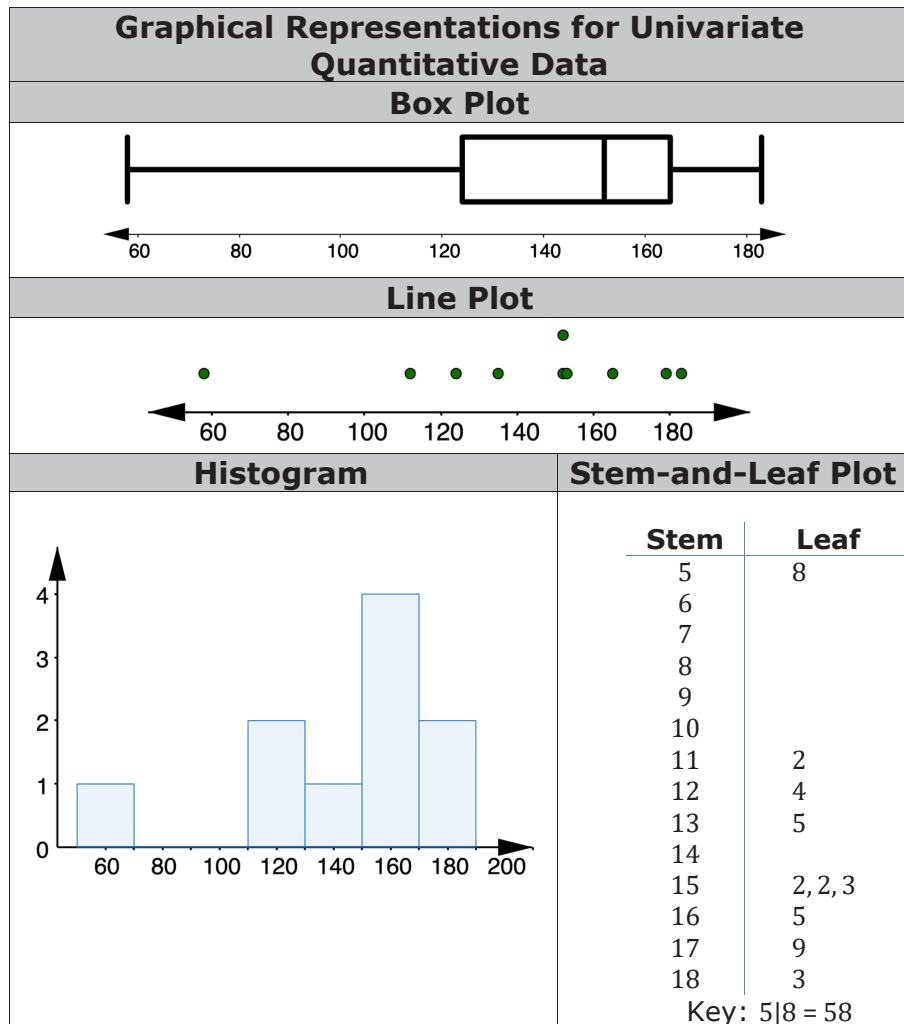
Possible Question	Possible Responses	Numerical or Categorical
What is your favorite color?	red, orange, yellow, green	☒ categorical ☐ numerical
How many people are in your family?	2, 3, 4, 5 ...	☐ categorical ☒ numerical
How old are you?	11, 12, 13 ...	☐ categorical ☐ numerical
How tall are you?		☐ categorical ☐ numerical
What is your favorite sport?		☐ categorical ☐ numerical
	watermelon, apple, pear, orange	☐ categorical ☐ numerical



12.1.2 Refresher: Choosing the Best Graphical Representation for Univariate Numerical Data

Four graphical representations that can be used to display **numerical data** that is **univariate** are box plots, histograms, line plots, and stem-and-leaf plots. Each of these graphical representations is shown in the table.

Univariate data is data that measures one characteristic of a population.



Work with your partner to complete the following.

- What do you notice about the displays?
- Complete the table by checking the displays that match each attribute.

Attribute	Box Plot	Line Plot	Histogram	Stem and Leaf Plot
Shows the distribution of the data set	☐	☐	☐	☐
Shows the shape of the data distribution	☐	☐	☐	☐
Data is displayed in intervals (also called bins)	☐	☐	☐	☐
Displays each individual data value	☐	☐	☐	☐
Displays a 5-number summary of the data set	☐	☐	☐	☐

3. Camilla wants to create a graphical display that represents the typing speeds of the 425 students at her school.

a. Which display(s) will best display the data?

- ☐ box plot
- ☐ line plot
- ☐ histogram
- ☐ stem-and-leaf plot

b. Explain your reasoning.

c. Camilla says, "Making a line plot of this data would take a lot of time and it would be hard to make sure each dot is in the right location on the number line, since speed can be given in fractions of seconds. I'd have to make 425 dots to represent each student!" Explain whether you agree with Camilla that a line plot would not best display the data.

4. Dallas records the highest temperature from each month over the past year in Immokalee, Florida.

$\{76, 78, 81, 85, 90, 92, 92, 93, 91, 87, 82, 78\}$

a. Which display(s) will best display the data?

- ☐ box plot
- ☐ line plot
- ☐ histogram
- ☐ stem-and-leaf plot

b. Explain your reasoning.

c. Dallas decides to change his data set to include the average weekly high temperature. Explain whether this changes the display he should use.

5. Basil recorded the weights, rounded to the nearest pound, of 15 dogs at a pet show.

7, 10, 12, 22, 28, 59, 62, 65, 72, 79, 85, 98, 100, 150, 163

- a. Explain which display would be better for the data, a line plot or a stem-and-leaf plot.

- b. Basil also recorded the weights, rounded to the nearest pound, of 15 cats at the pet show.

4, 4, 5, 7, 7, 8, 8, 8, 9, 10, 11, 12, 13, 13, 14

Explain which display would be better for the data, a line plot or a histogram.

- c. He updates his data to show the weights to the nearest tenth.

4.1, 4.3, 5.2, 7.6, 7.9, 8.0, 8.2, 8.7, 9.3, 10.5, 11.2, 12.8, 13.4, 13.5, 14.6

Explain which display would be better for the data, a line plot or a histogram. If your recommendation is different from Part B, explain why.

6. Ms. Rallen asked her students how many pets they each have, and wants to use a graphical representation that will allow her students to see the overall shape of the data and give them enough information to calculate the 5-number summary.

- a. Which graphical representations should Ms. Rallen consider?

- ☐ box plot
- ☐ line plot
- ☐ histogram
- ☐ stem-and-leaf plot

- b. Explain your reasoning.



12.1.3 Collaboration: Graphical Representations of Univariate Categorical Data

Graphical representations that can be used to display univariate categorical data include **bar graphs**, **circle graphs**, **line plots**, **frequency tables**, and **relative frequency tables**.

Categorical data is often collected in a frequency table before creating another display.

Consider the data summarized in the frequency table and relative frequency table below, collected from a survey on how students get to school.

Frequency Table

Mode of Transportation	Frequency
Bus	129
Car	71
Bike	65
Walk	35

Relative Frequency Table

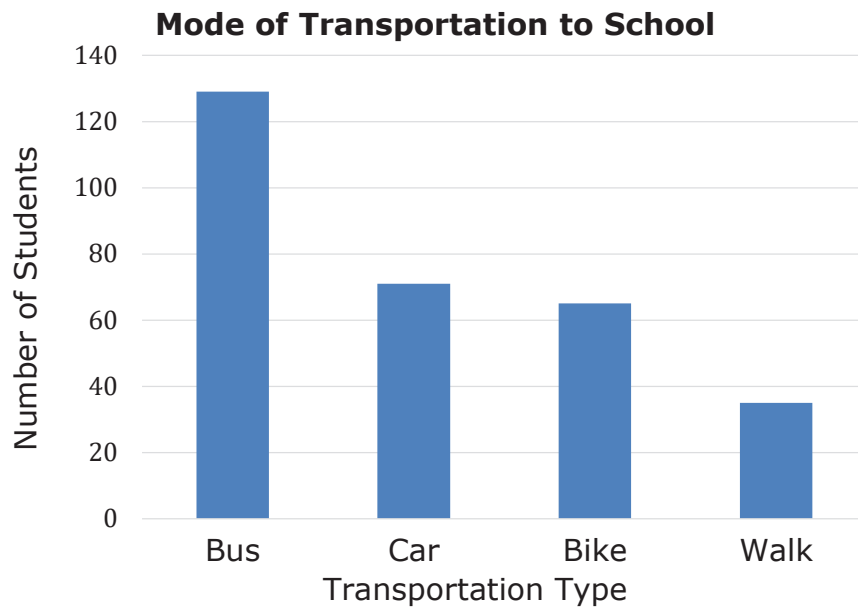
Mode of Transportation	Relative Frequency
Bus	43%
Car	24%
Bike	22%
Walk	11%

1. Summarize the similarities and differences between the two displays.

Similarities	Differences

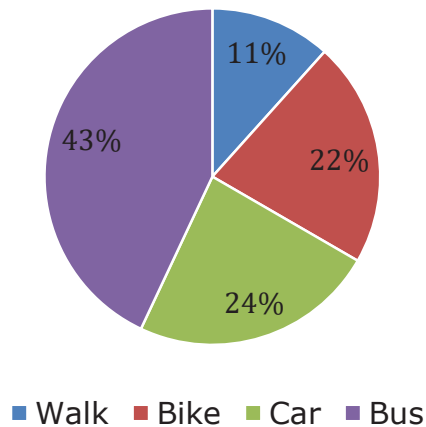
2. How was the data from the frequency table used to construct the relative frequency table?

The vertical bar graph and circle graph show the same data.



The circle graph shows the same data.

Mode of Transportation to School



3. Complete the statements.

The bar graph visually shows

- ☐ how each part relates to the whole.
- ☐ how each part compares to the other parts.

The circle graph shows

- ☐ how each part relates to the whole.
- ☐ how each part compares to the other parts.

The data from the frequency table was used to

construct the

- ☐ bar graph.
- ☐ circle graph.

The data from the relative frequency table was used to

construct the

- ☐ bar graph.
- ☐ circle graph.

4. Similar to a line plot (dot plot) for quantitative data, a line plot can also be used for categorical data by listing each category on the x -axis. Would a line plot be a good choice for displaying this data? Explain your reasoning.



12.1.4 Wrap-Up: Cool Down

1. My favorite graphical representation to display

numerical univariate data is

- box plot.
- line plot.
- histogram.
- stem and leaf.

It is my favorite because ... _____

2. My favorite graphical representation to display

univariate categorical data is a

- bar graph.
- circle graph.
- line plot.
- frequency table.

It is my favorite because ... _____



Unit 12, Lesson 2: Creating Displays for Numerical Data Sets



Benchmarks

MA.912.DP.1.1 Given a set of data, select an appropriate method to represent the data, depending on whether it is numerical or categorical data and on whether it is univariate or bivariate.

MA.912.DP.1.2 Interpret data distributions represented in various ways. State whether the data is numerical or categorical, and whether it is univariate or bivariate and interpret the different components and quantities in the display.



Learning Targets

- ☐ I can distinguish between univariate and bivariate data, including which representations can be used to display each type of data set.
- ☐ I can create a graphical display for univariate numerical data.
- ☐ I can create a graphical display for bivariate numerical data.